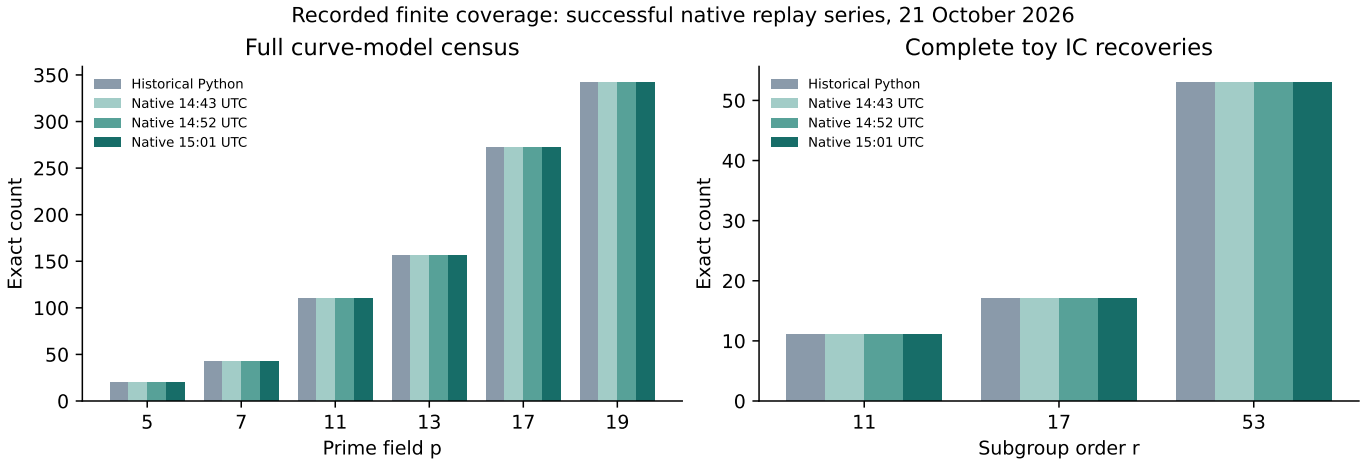
Quadratic twists: native replay and checked implications

Reproducible correctness study

21 October 2026

Outcome. The complete finite workload now executes in C++17. All 434 deterministic output fields agree with the frozen historical results, including newly collected IC matrices and their attempt counts. The historical Python archive is unchanged. Eighteen Lean statements pass the repository proof gate, with zero trusted native proofs. Universal geometric bridges remain partial; no general hardness or speedup conclusion is drawn.



Recorded counters from the successful 14:43, 14:52 and 15:01 UTC native runs. The earlier port attempt failed parity comparison and has no accepted result. Exact census counts; no statistical error bar or extrapolation. The host is uncontrolled Linux x86-64, GCC 14.2.0 and OpenSSL 3.5.7. UTC boundaries are provenance, not admitted algorithm timings.

Replayed workload	Recorded count
Nonsingular models, primes 5, 7, 11, 13, 17, 19	942
Ordered S3 triples	4,188,162
Ordered S4 quadruples, primes 5 and 7	113,342
Extension points	260,064
Twist group-law identities	289,440
Scalar transport checks	94,176
Common odd-prime cases	274
Complete toy IC logarithms	81
Bilinearity identities	3,219

All four trace-zero fixtures, the subfield example and four transported relation histograms match. The original 12 seeded associativity samples per model outside prime 5 remain samples, not exhaustive proofs. Process timing, source provenance and SymPy version fields are excluded from mathematical parity. C++ integer-seeded MT19937 reproduces the original integer draws; exact sparse polynomial expansion over $\mathbb{Z}$ replaces SymPy.

Independent verification and controls

The separate tuple-based auditor verifies 84 newly emitted IC matrices: three headline runs and the 81 all-log runs. It independently recomputes the pairings, field logarithms, scalars and ranks. Corrupting a new relation or removing a log certificate is rejected. Changing the historical census counter is also rejected. Existing source-corruption and false historical-relation controls remain.

Before registering the subfield target in a simple polynomial basis, the new basis checker verifies 2,401 elements, 2,475 points and 2,025 group-map identities. Here $u^2 = 3$, $v^2 = 1 + u$, $w = v$, $u = w^2 - 1$, and $w^4 + 5w^2 + 5 = 0$. The model catalog has 1,921 exact ICV1 models and 27 selected EC1 subgroup representations. An unselected generator is left unknown. The normalization of $cy^2 = x^3 + ax + b$ is explicit: $(x, y) \mapsto (cx, c^2y)$ on the monic model with coefficients c^2a, c^3b. No base-field log transfer is inferred from a shared model or j-invariant.

Proofs and experimentally discriminated claims

The Lean module proves the mixed-relation implication: if an additive endomorphism fixes A and $A + B$ and negates B, then $2B = 0$. Coprime annihilators force $B = 0$, proving the odd-order consequence. The finite coprime-order homomorphism theorem, trace/anti-trace identities, common odd-order divisor implication, three ring identities, affine twist-map and monic-normalization equations, and algebraic embedding-degree-two implication are also checked. All hypotheses are explicit in the proof map. Four further statements use the actual elliptic point group, the q-power Frobenius map and rational base-change homomorphism. Rational lifts are proved fixed; a Frobenius-negated mixed contribution is proved killed by two and zero under an odd annihilator. The twist-to-negation identification remains explicit.

S3 symmetry	`IDC1h07a9b2e6b0a3a9d3`
Vieta cancellation	`IDC1h98f781bdd4756596`
Twist equation	`IDC1hfb8edf881dcde851`

The records name the exact checked Lean source. Native certificate replay uses 32 derived points and 16 extra points and rejects false identities. Point evaluation alone is weaker than the accompanying universal ring proofs.

The valid-point census retains a nonzero 2-torsion exception. S3/S4 conditions are preserved under the geometric twist map, and the matched transported relation histograms agree in four fixtures. All 81 pairing/IC logarithms are recovered. A smooth or anomalous twist does not force the same original order or a nonzero rational homomorphic transfer. The subfield fixture's order-three target remains outside the smaller rational group. The restricted relation family has rank 5 of 6 despite 272 smooth relations from 2,448 candidates; a broader seeded family reaches full rank. These findings have their exact finite scope, not an asymptotic interpretation.

Remaining obligations and provenance

Twist embedding and its Frobenius negation property, full rational-eigenspace identification, general summation-polynomial lifting, Weil-pairing nondegeneracy and the full reduction, and universal trace-zero geometry remain open bridges. General GHS construction and general ECDLP hardness are not established. Invalid-curve attacks are outside scope. Conductor's shared tool and task state remain unavailable. Rust and Lean are now installed; current repository checks are recorded separately in `validation/`.

The original archive, import report, native source snapshots, failed first port comparison and failed first proof build remain available. No benchmark or leaderboard row or IC/rho cost graph is promoted. This new graph extends study coverage; earlier reports stay attached to their original records.

Proof map and reproduction

Run record	Observed outcome
`port-attempt-1`	Rejected: parity comparator incorrectly treated two JSON nulls as unequal. Log and exact source retained; no accepted receipt.
`run-20261021T144307Z`	434 fields match; independent audit verifies 84 IC certificates.
`run-20261021T145210Z`	Same parity and audit; tower/simple-basis certificate added.
`run-20261021T150125Z`	Final source snapshot; same parity, audit and basis checks; final negative controls pass.

The first 14 declarations are in `TwistProof`, with four further declarations in `FrobeniusBridge`. Each statement's exact hypotheses and the remaining geometric bridge are documented in `lean/README.md`. The proof gate audits all 18 declarations.

Lean declaration	Proved implication
`mixed_relation_two_torsion`	Fixed A, negated B, fixed $A+B$ imply $2B=0$.
`coprime_annihilators`	Two coprime annihilators force zero.
`mixed_relation_odd`	An odd annihilator removes the 2-torsion exception.
`finite_hom_coprime`	A homomorphism across coprime finite orders is zero.
`trace_antitrace_sum`	Trace plus anti-trace is $2P$.
`trace_fixed`	An involution fixes its trace.
`antitrace_negated`	An involution negates its anti-trace.
`common_order_divides_q_plus_one`	An odd common order divisor divides $q+1$.
`symmetric_s3`	The two S3 ring expressions agree.
`vieta_twist_cancellation`	The parameter cancels in the Vieta expression.
`twist_equation_identity`	The cleared equation is a multiple of u^2-d.
`twist_affine_map`	The x-fixing map satisfies the untwisted equation.
`monic_normalization`	The stated scaling satisfies the monic equation.
`embedding_degree_two`	$q=-1$ and $2\neq 0$ imply $q^2=1$, $q\neq 1$.
`FrobeniusBridge`: actual elliptic-curve points	
`frobenius_fixes_rational`	Actual q-power Frobenius fixes rational base-change points.
`rational_mixed_two_torsion`	A rational relation with Frobenius-negated B gives $2B=0$.
`rational_mixed_odd`	With an odd annihilator, that B is zero.
`rational_relation_has_zero_twist_term`	$B=0$ and the two lifted rational points coincide.

The group/ring implications and stated actual-point theorems are fully proved. Identifying the embedded twist points with the minus eigenspace, and closing the other geometric bridges, remain explicit obligations.

```
study=research/quadratic_twist_evidence_20261021
bash "$study/run-native.sh" /tmp/twist-native-fresh
bash "$study/test-native.sh" /tmp/twist-native-fresh /tmp/controls-fresh
tools/lean/proof-gate.sh --out /tmp/proofs "$study/lean"
```

The adjacent Markdown report links all native records, controls, certificates, proof receipts and the requirement map. Primary references remain in the preserved historical report: Semaev (ePrint 2004/031), Gorla–Massierer (arXiv 1405.1059), Menezes–Okamoto–Vanstone, and Gaudry–Hess–Smart.